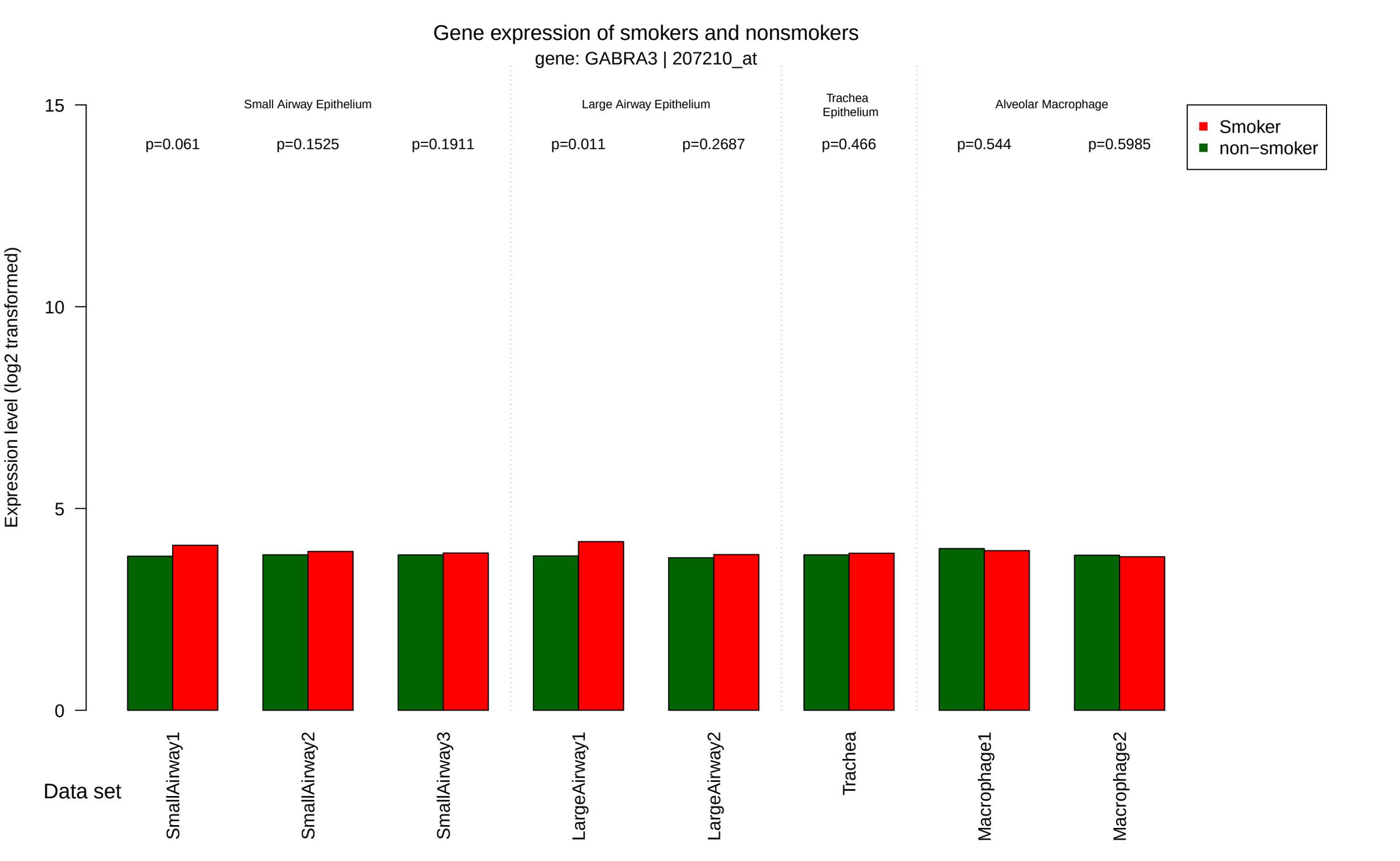
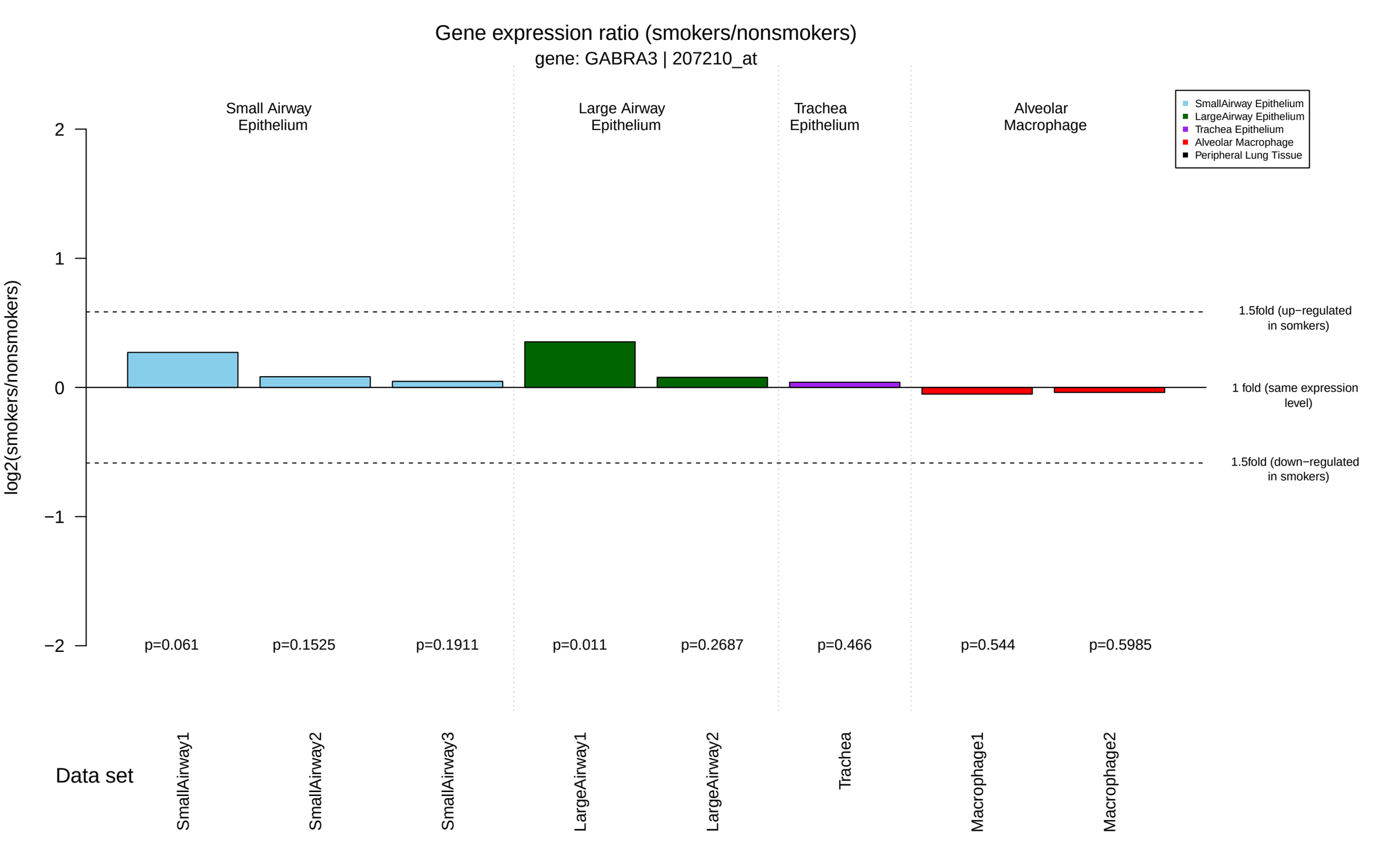
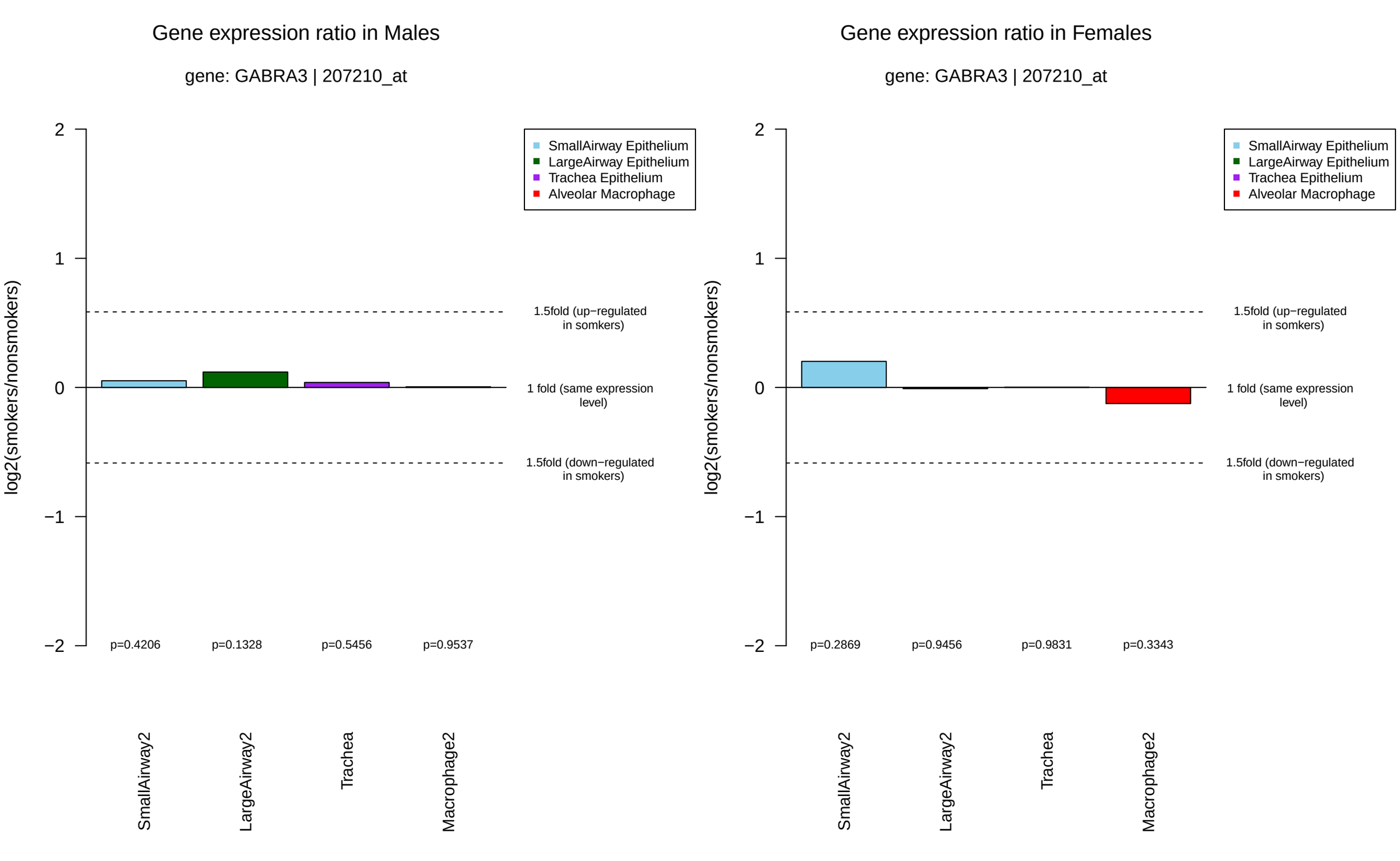
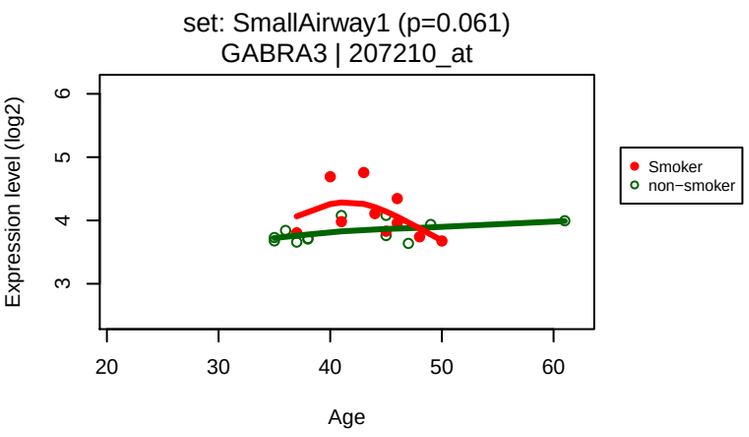
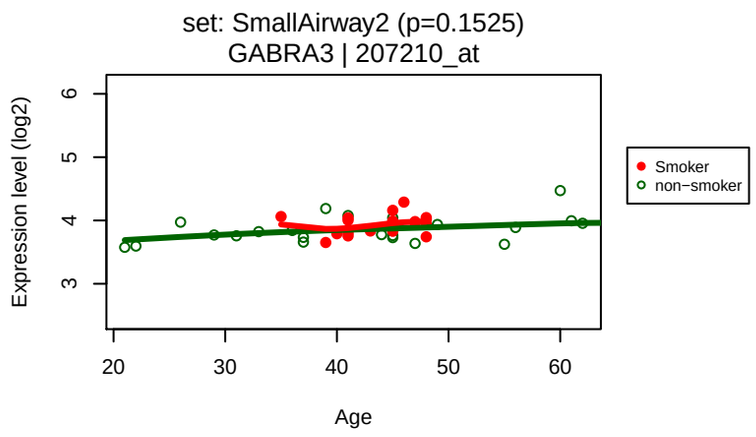
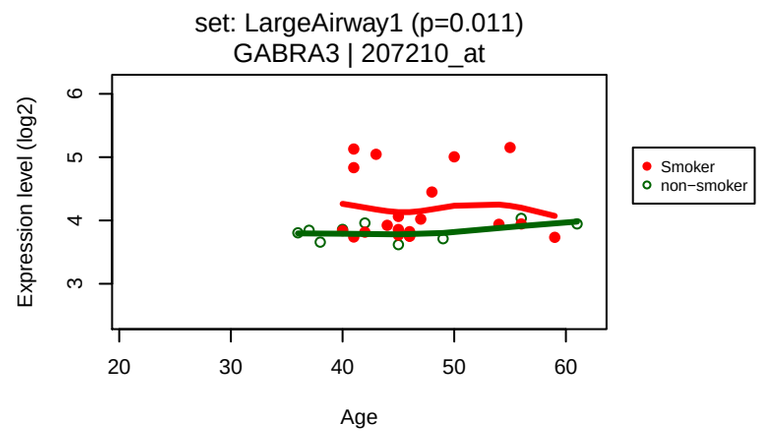

Gene expression of smokers and nonsmokers
gene: GABRA3 | 207210_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.061
p=0.1525
p=0.1911
p=0.011
p=0.2687
p=0.466
p=0.544
p=0.5985
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

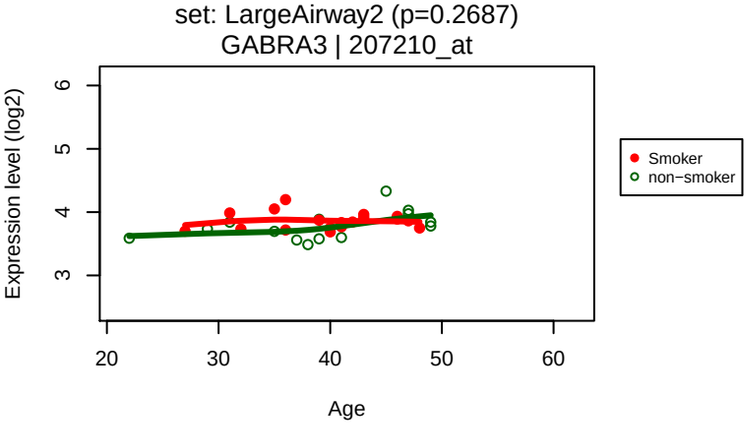

Gene expression ratio (smokers/nonsmokers)
gene: GABRA3 | 207210_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.061
p=0.1525
p=0.1911
p=0.011
p=0.2687
p=0.466
p=0.544
p=0.5985
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

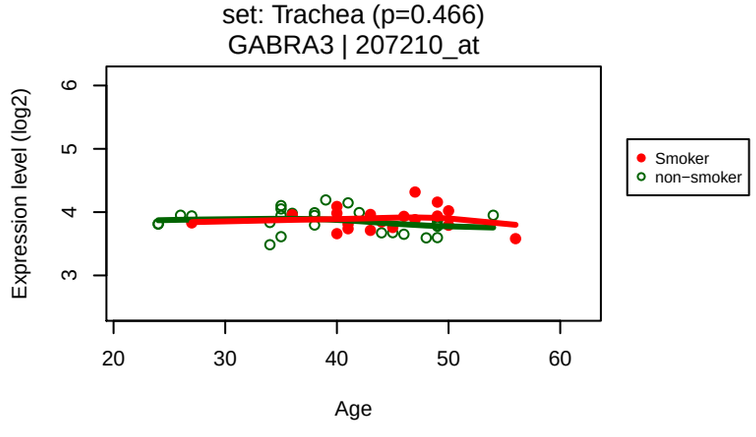

Gene expression ratio in Males
gene: GABRA3 | 207210_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4206
p=0.1328
p=0.5456
p=0.9537
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GABRA3 | 207210_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2869
p=0.9456
p=0.9831
p=0.3343
SmallAirway2
LargeAirway2
Trachea
Macrophage2

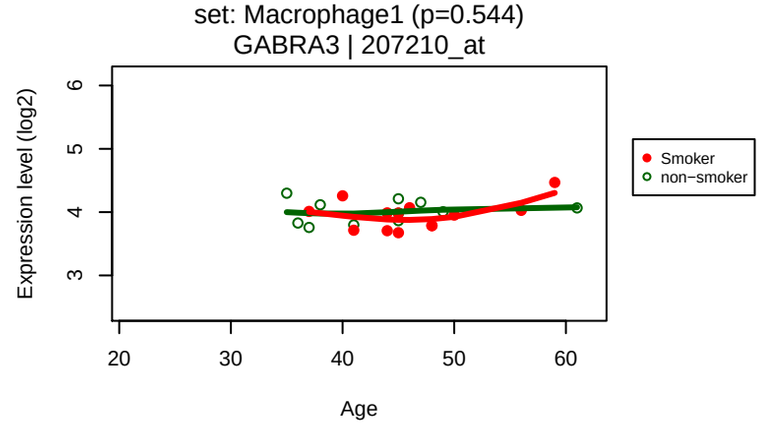

set: SmallAirway1 (p=0.061)
GABRA3 | 207210_at
Expression level (log2)
Age
Smoker
non−smoker

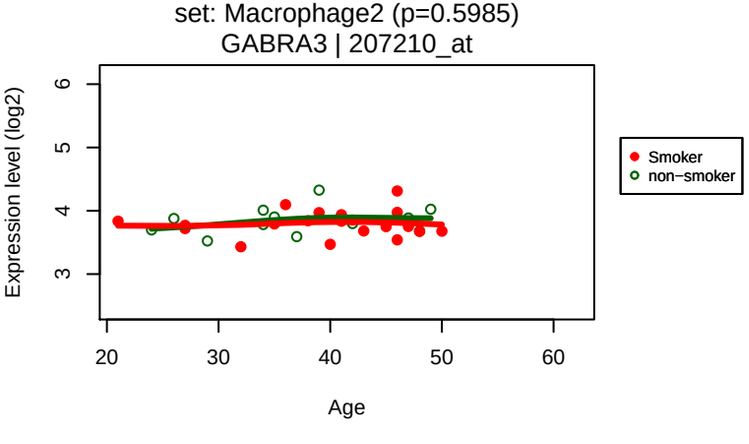

set: SmallAirway2 (p=0.1525)
GABRA3 | 207210_at
Expression level (log2)
Age
Smoker
non−smoker

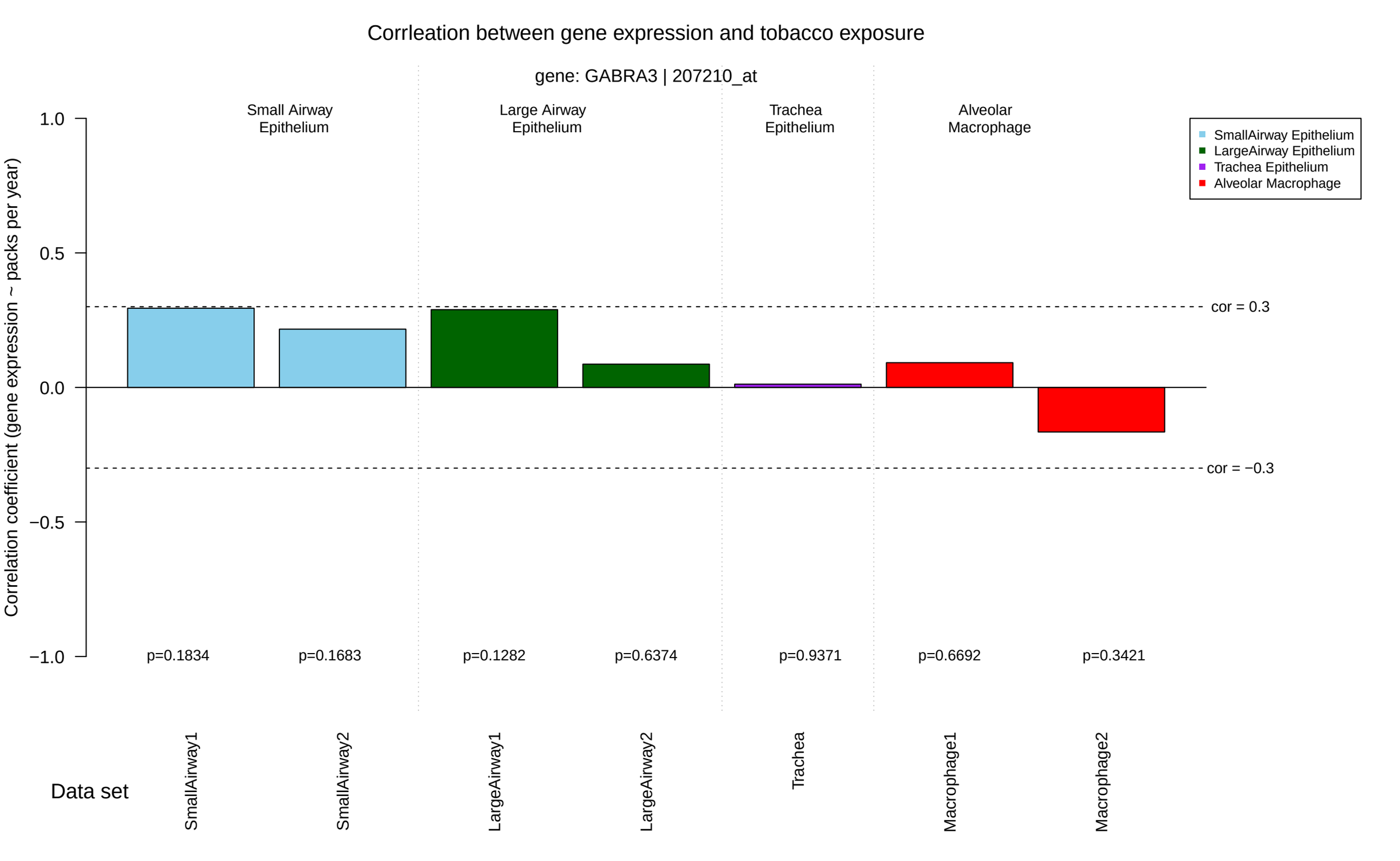

set: LargeAirway1 (p=0.011)
GABRA3 | 207210_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2687)
GABRA3 | 207210_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.466)
GABRA3 | 207210_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.544)
GABRA3 | 207210_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5985)
GABRA3 | 207210_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GABRA3 | 207210_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1834
p=0.1683
p=0.1282
p=0.6374
p=0.9371
p=0.6692
p=0.3421
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2